CAREERINTELL AI | *Comprehensive Project Files & Architecture Guide*

Technical reference explaining the purpose, architecture, and exact use case of every file in the project repository for academic evaluation and development.

Project: CareerIntell AI (T.Y Project)	**Frontend:** React 18 + Vite + TS + Tailwind	**Backend:** FastAPI Python 3.14 + PyMuPDF	**Database:** SQLite (Async SQLAlchemy)

1. Core Frontend Application (src/)

SRC/ CORE FILES & COMPONENTS	
src/pages/RoadmapPage.tsx	**Role:** Personalized learning pathway & career milestones generator view. ***CareerIntell Use Case:*** *Visualizes step-by-step learning phases, milestones, curated tutorials/videos, and completion tracking for skills identified as gaps during resume/JD matching.*
src/types/	**Role:** Global TypeScript interface definitions and data contract schemas. ***CareerIntell Use Case:*** *Maintains type safety across the frontend for objects like User, StudentProfile, Skill, ExtractedSkill, SkillGapItem, Roadmap, and InterviewSession.*
src/utils/	**Role:** Frontend utility functions and formatters. ***CareerIntell Use Case:*** *Provides reusable logic such as Tailwind CSS class merging (clsx/cn), date formatters, currency conversions (USD/INR), and auth token extraction.*
src/App.tsx	**Role:** Root React component, global router, and route guard protection. ***CareerIntell Use Case:*** *Configures React Router paths (Landing, Dashboard, Analyze, Roadmap, Interviews, Assessments, TPO Portal) and handles role-based student/TPO permissions.*
src/index.css	**Role:** Global stylesheet combining Tailwind directives with custom design tokens. ***CareerIntell Use Case:*** *Implements the Frosted Pearl & Ambient Slate aesthetic (.pro-card, .pro-panel, translucent blur backdrops, radial pearl glows, and custom scrollbars).*
src/main.tsx	**Role:** Application DOM entry point and React root mounting script. ***CareerIntell Use Case:*** *Bootstraps the React virtual DOM into inside index.html, wrapping the application in AuthProvider, ThemeProvider, and BrowserRouter.*
src/vite-env.d.ts	**Role:** TypeScript type declarations for Vite environment features. ***CareerIntell Use Case:*** *Allows developers to access Vite environment variables (like import.meta.env.VITE_API_URL) without TypeScript compilation errors.*

2. HTML Containers & Access Hubs

HTML & BROWSER ENTRY FILES	
index.html	**Role:** Single Page Application (SPA) HTML container for browser delivery. ***CareerIntell Use Case:*** *Supplies the viewport, loads Google Fonts (Outfit & Plus Jakarta Sans), and mounts the bundled React UI into the #root DOM element.*
launch.html	**Role:** Offline Project Access Hub & Quick Reference Dashboard. ***CareerIntell Use Case:*** *A standalone, zero-dependency HTML dashboard allowing faculty or reviewers to launch live app links, open Swagger API docs, and view architecture diagrams.*

3. Server Automation Scripts

ONE-CLICK AUTOMATION	
start_app.bat	**Role:** One-click Windows batch startup script for full system execution. ***CareerIntell Use Case:*** *Launches the FastAPI Python backend on port 8000 and the Vite frontend on port 5173 simultaneously, then automatically launches http://localhost:5173 in browser.*

4. Build Tooling & Configuration Pipeline

BUILD & TRANSPILATION TOOLING	
package.json	**Role:** Node.js package manifest specifying libraries, scripts, and versions. ***CareerIntell Use Case:*** *Declares dependencies (React 18, Lucide React, Axios, Canvas Confetti) and CLI scripts (npm run dev, npm run build, npm run preview).*
package-lock.json	**Role:** Exact dependency tree lockfile ensuring reproducible installations. ***CareerIntell Use Case:*** *Guarantees that identical library versions and checksums are installed on every machine, preventing cross-environment package drift.*
vite.config.ts	**Role:** Vite bundler configuration, HMR engine, and API reverse proxy. ***CareerIntell Use Case:*** *Configures lightning-fast Hot Module Replacement (HMR), React compiler plugins, and proxies /api requests to http://127.0.0.1:8000.*
tailwind.config.js	**Role:** Tailwind CSS theme customization and design token extensions. ***CareerIntell Use Case:*** *Defines the Frosted Pearl palette (slate-50 to slate-900), custom frosted glass opacities, typography families, and elevation drop-shadows.*
postcss.config.js	**Role:** CSS post-processing pipeline configuration. ***CareerIntell Use Case:*** *Connects Tailwind CSS and Autoprefixer into Vite's build lifecycle to compile modern CSS with complete cross-browser vendor prefixing.*
tsconfig.json / tsconfig.app.json / tsconfig.node.json	**Role:** TypeScript compiler configurations managing type validation. ***CareerIntell Use Case:*** *Enforces strict type safety across React source files (tsconfig.app.json) and Node configuration files (tsconfig.node.json).*
***.tsbuildinfo**	**Role:** TypeScript compiler incremental build caches. ***CareerIntell Use Case:*** *Saves internal dependency graphs so subsequent tsc -b checks only analyze modified files, speeding up production bundle builds.*

5. Project Documentation & Version Control

DOCUMENTATION & VCS ARTIFACTS	
.gitignore	**Role:** Git version control exclusion rules. ***CareerIntell Use Case:*** *Prevents temporary, sensitive, or heavy directories from being committed to Git (e.g. node_modules/, dist/, .env, __pycache__/, .db).*
DEVELOPMENT_STATUS.md	**Role:** Active engineering tracker and college project milestone record. ***CareerIntell Use Case:*** *Tracks completed components (Auth, PyMuPDF NLP parser, skill gap diagnostics, TPO authentication) and remaining feature deliverables for evaluation.*
FEATURE_GAP_MATRIX.md	**Role:** Architectural comparison matrix against industry benchmarks. ***CareerIntell Use Case:*** *Contrasts CareerIntell's semantic intelligence against traditional placement portals (highlighting PyMuPDF layout parsing and equivalence clusters).*
PROJECT_AUDIT.md	**Role:** Comprehensive security, database, and endpoint audit report. ***CareerIntell Use Case:*** *Documents database schemas, JWT token lifecycle, password hashing algorithms, and REST endpoint contracts for academic documentation.*

6. End-to-End System Execution Flow

Step	File / Component	Action Performed in CareerIntell Flow
1. Launch	**start_app.bat**	Launches FastAPI on port 8000 and Vite on port 5173 concurrently, then opens browser.
2. Boot	**index.html → main.tsx**	Browser loads Outfit & Plus Jakarta fonts; React mounts <App /> into DOM.
3. Styling	**index.css + tailwind.config.js**	Applies Frosted Pearl & Ambient Slate aesthetic with translucent card backdrops.
4. Route	**App.tsx**	Routes user to Landing Page, Student Dashboard, Analyze Page, or TPO Portal.
5. Diagnose	**AnalyzePage.tsx → vite.config.ts**	Proxies resume upload to FastAPI PyMuPDF parser, rendering 99% match & 0 false gaps.
6. Learn	**RoadmapPage.tsx**	Renders personalized phase-by-phase learning curriculum for identified skill gaps.